Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/28/2006
PI Name:	Buckmaster, John Ph.D.		
Project Title:	Smolder-edge-waves and Edge-flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	61801	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	01/01/2001	**End Date:**	11/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Hicks, Michael C.	**Contact Phone:**	216-433-6576
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2516		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The study of edge-flames in contexts relevant to the microgravity experimental program, with particular efforts to describe edge-flames in smolder combustion, an issue of great importance in space safety. To provide, for the first time, a description of edge-waves in smolder combustion, paradigms for 2-dimensional smolder waves with edges. Most of our knowledge of smolder is either 1-dimensional, or 2-dimensional in a boundary-layer like context. Our understanding of smolder spread in a two-dimensional context outside of the aforementioned contexts is inadequate, and so our knowledge of how smolder can spread or transition to flaming combustion is incomplete, dangersously so.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:			

Task Progress:	Recently we have examined flames in narrow tubes, and have submitted the following to the 2006 Combustion Institute Proceedings: 'Flames in Narrow Circular Tubes' T.L.Jackson, J.Buckmaster, Z.Lu, D.C.Kyritsis, L.Massa We examine an axi-symmetric deflagration located in a tube with thermally active walls. It is noted that the flame-in-tube configuration defines a classical edge-flame, a flame in a shear flow for which there is a water-shed solution for a critical value of the Damkohler number, ignition front solutions for larger values of the Damkohler, and failure wave solutions for smaller values. We examine semi-infinite tubes with a heat flux imposed at the tube wall ends, to simulate experiments reported in the 30th Symposium. We identify parameters for which stable solutions are obtained at certain flow rates, but unstable solutions are generated at higher flow rates, followed by stable solutions at still higher flow rates. These solutions are consistent with the experimental record. Instability leads either to regular oscillations or to a violent process characterized by quenching and re-ignition.
Bibliography Type:	Description: (Last Updated: 04/21/2021)
Articles in Peer-reviewed Journals	Chen M, Buckmaster J. "Modelling of combustion and heat transfer in 'Swiss roll' micro-scale combustors." Combustion Theory and Modelling, 2004 Dec;8(4):701-20. http://dx.doi.org/10.1088/1364-7830/8/4/003 , Dec-2004
Articles in Peer-reviewed Journals	Liu Y, Chen M, Buckmaster J, Jackson TL. "Smolder waves, smolder spots, and the genesis of tribrachial structures in smolder combustion." Proceedings of the Combustion Institute, 2005 Jan;30(1):323-9. http://dx.doi.org/10.1016/j.proci.2004.08.183 , Jan-2005
Articles in Peer-reviewed Journals	Jackson TJ, Buckmaster J, Lu Z, Kyritsis DC, Massa L. "Flames in narrow circular tubes." Proceedings of the Combustion Institute, 2007 Jan;31(1):955-62. http://dx.doi.org/10.1016/j.proci.2006.07.032 , Jan-2007
Articles in Peer-reviewed Journals	Lu Z, Buckmaster J, Chen M, Massa L. "Instabilities of reverse smolder waves." Combustion Theory and Modeling. 2006;10(3):515-34. https://doi.org/10.1080/13647830500497682 , Dec-2006
Articles in Peer-reviewed Journals	Kessler DA, Short M, Buckmaster J. "Edge-flames and cellular structures in oscillatory premixed counterflows." Proceedings of the Combustion Institute. 2005 Jan;30(1):331-7. https://doi.org/10.1016/j.proci.2004.08.025 , Jan-2005